

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459217.6514 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.01
Transit depth (Rp/Rs)^2	2.83 +/- 0.35 [%]
Orbital Inclination (inc)	83.74 +/- 0.27
Airmass coefficient 1 (a1)	1.2202 +/- 0.0073
Airmass coefficient 2 (a2)	-0.1179 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	0.59 %
Best Comparison Star	None
Optimal Aperture	5.23
Optimal Annulus	12.74
Transit Duration (day)	0.0707 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.01 vs 0.1432 (deviation 2.48 σ)
Duration fit vs published	0.0707 d vs 0.0604 d (deviation 2.57 σ)
Mid-time O-C	-0.6 min
Depth SNR	16.8
Residual scatter	0.59 %

Light curve

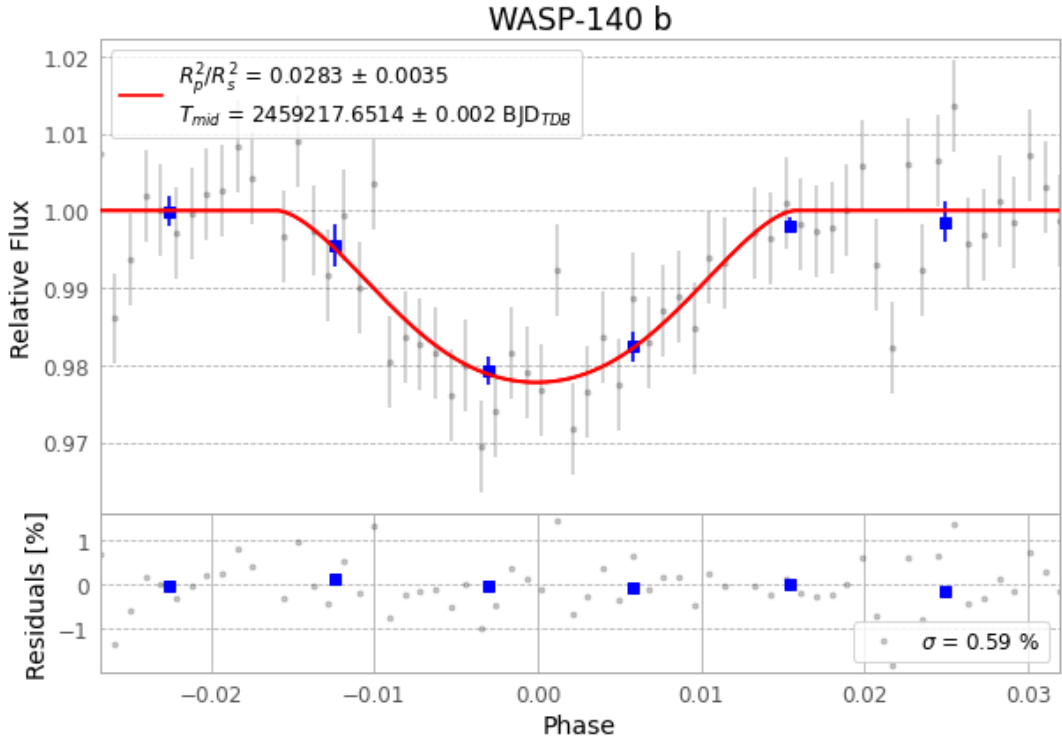


Figure 1 — FinalLightCurve_WASP-140 b_2021-01-02.png (SHA-256 0030280d8a5f7802...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [246, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c3f497e8aae5a6a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-140210103020612.FITS → WASP-140210103051512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-210103.log (dc49bb2c11e8...), FinalLightCurve_WASP-140 b_2021-01-02.png (0030280d8a5f...), FinalLightCurve_WASP-140 b_2021-01-02.csv (555a8769a965...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:31Z.